K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

LINUX BASH SCRIPTING LAB

(ENCS356)

Lab file Submitted to

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For

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In

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(Cyber Security)

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Experiment No. 1

Exploring Linux System Commands, Run Levels (Targets), and Help Utilities.

Problem Statement

Efficient use of the Linux operating system requires a strong understanding of basic system commands, system states, and built-in help mechanisms. Unlike graphical operating systems, Linux relies heavily on command-line utilities for system interaction, administration, and automation.

The problem is to explore and practice essential Linux system commands, understand system run levels / targets, and effectively use Linux help utilities such as man, *--help*, and *auto*- completion to access command documentation and usage information.

Aim

To study and execute basic Linux system commands, understand Linux run levels (systemd targets), and use Linux help utilities such as `man`, `--help`, and tab auto-completion.

Theory

Linux Command Line Interface (CLI)

The Linux Command Line Interface allows users to interact with the operating system by typing commands. It provides better control, flexibility, and efficiency compared to graphical interfaces, especially for system administration and automation.

System Run Levels / Targets

Modern Linux systems use **systemd**, where traditional run levels are replaced by **targets**.

Common systemd targets:

- `graphical.target` – Multi-user mode with GUI
- `multi-user.target` – Multi-user mode without GUI
- `rescue.target` – Single-user rescue mode
- `poweroff.target` – Shutdown system
- `reboot.target` – Reboot system

Linux Help Utilities

Linux provides built-in help tools for learning commands:

- man – Displays complete manual pages
- --help – Shows brief command usage
- **Tab auto-completion** – Completes commands and file names automatically

Commands Used

Commands	Description
pwd	Display present working directory
ls	List files and directories
cd	Change directory
whoami	Display current user
date	Show system date and time
uptime	Show system running time
man	Display manual pages
--help	Display help for a command
systemctl	Manage system services and targets
runlevel	Display previous and current run level
clear	Clear terminal screen

Procedure / Steps

Step 1: Login: Log in to the Linux virtual machine and open the **Terminal**.

Step 2: Execute Basic Linux Commands in the Terminal.

Commands:

- pwd shows current directory
- ls lists files
- whoami displays logged-in user
- date shows system date/time
- uptime shows how long system is running

```
erin@erin-VirtualBox: ~
erin@erin-VirtualBox:~$ pwd
/home/erin
erin@erin-VirtualBox:~$ ls
Desktop  Documents  Downloads  erin  lab_work  Music  Pictures  Public  snap  Templates  Videos
erin@erin-VirtualBox:~$ whoami
erin
erin@erin-VirtualBox:~$ date
Mon Mar  2 01:46:59 PM IST 2026
erin@erin-VirtualBox:~$ uptime
 13:47:08 up 5 min,  1 user,  load average: 0.43, 1.38, 0.81
erin@erin-VirtualBox:~$
```

Step 3: Directory Navigation using:

- cd /home moves to home directory.
- cd ~ returns to user's home directory.

```
erin@erin-VirtualBox: ~
erin@erin-VirtualBox:~$ cd /home
erin@erin-VirtualBox:/home$ cd ~
erin@erin-VirtualBox:~$
```

Step 4: Using Manual Pages

- man ls
- man cd

```
erin@erin-VirtualBox: ~
LS(1)                                User Commands                                LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List  information  about the FILEs (the current directory by default).  Sort entries alphabetically if none of
       -cftuvSUX nor --sort is specified.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

       -b, --escape
              print C-style escapes for nongraphic characters

       --block-size=SIZE
              with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

       -B, --ignore-backups
              do not list implied entries ending with ~
 Manual page ls(1) line 1 (press h for help or q to quit)
```

```
erin@erin-VirtualBox: ~
erin@erin-VirtualBox:~$ cd /home
erin@erin-VirtualBox:/home$ cd ~
erin@erin-VirtualBox:~$ man ls
erin@erin-VirtualBox:~$ man cd
No manual entry for cd
erin@erin-VirtualBox:~$
```

Step 5: Using Help Option:

- ls -help
- systemctl --help

```
erin@erin-VirtualBox:~$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
      --block-size=SIZE      with -l, scale sizes by SIZE when printing them;
                             e.g., '--block-size=M'; see SIZE format below

  -B, --ignore-backups       do not list implied entries ending with ~
  -c                         with -lt: sort by, and show, ctime (time of last
                             change of file status information);
                             with -l: show ctime and sort by name;
                             otherwise: sort by ctime, newest first

  -C                         list entries by columns
      --color[=WHEN]         color the output WHEN; more info below
  -d, --directory            list directories themselves, not their contents
  -D, --dired                generate output designed for Emacs' dired mode
  -f                         list all entries in directory order
  -F, --classify[=WHEN]      append indicator (one of */=>@|) to entries WHEN
      --file-type            likewise, except do not append '*'
      --format=WORD          across -x, commas -m, horizontal -x, long -l,
                             single-column -1, verbose -l, vertical -C
```

```
erin@erin-VirtualBox:~$ systemctl --help
systemctl [OPTIONS...] COMMAND ...

Query or send control commands to the system manager.

Unit Commands:
  list-units [PATTERN...]             List units currently in memory
  list-automounts [PATTERN...]        List automount units currently in memory,
                                      ordered by path
  list-paths [PATTERN...]             List path units currently in memory,
                                      ordered by path
  list-sockets [PATTERN...]           List socket units currently in memory,
                                      ordered by address
  list-timers [PATTERN...]            List timer units currently in memory,
                                      ordered by next elapse
  is-active PATTERN...                Check whether units are active
  is-failed [PATTERN...]              Check whether units are failed or
                                      system is in degraded state
  status [PATTERN...|PID...]          Show runtime status of one or more units
  show [PATTERN...|JOB...]            Show properties of one or more
                                      units/jobs or the manager
  cat PATTERN...                      Show files and drop-ins of specified units
  help PATTERN...|PID...              Show manual for one or more units
  list-dependencies [UNIT...]         Recursively show units which are required
                                      or wanted by the units or by which those
                                      units are required or wanted
  start UNIT...                       Start (activate) one or more units
  stop UNIT...                        Stop (deactivate) one or more units
  reload UNIT...                      Reload one or more units
```

Step 6: Check System Run Level / Target

- runlevel-
- systemctl get-default

```
erin@erin-VirtualBox:~$ runlevel
N 5
erin@erin-VirtualBox:~$ systemctl get-default
graphical.target
erin@erin-VirtualBox:~$
```

Outcomes

- Successful execution of basic Linux commands
- Display of manual pages and help options
- Identification of current run level/target
- Improved command-line efficiency

Result

Thus, basic Linux system commands, system run levels and help utilities were successfully explored. The student gained practical understanding of Linux command-line interaction and Linux documentation tools.